

Automated Diagnosis of Alzheimer's Disease Using Medical Imaging

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Abstract—Alzheimer's disease is the most common cause of dementia, and early detection is essential. This study investigates MRI-based automated diagnosis using both traditional and deep learning methods. Radiomic features were extracted and used with XGBoost and SVM, while Convolutional Neural Networks (CNNs) including EfficientNetB0 were trained directly on image data. EfficientNetB0 achieved the highest test accuracy (96.56%), showing that lightweight deep learning models are promising tools for medical image classification.

Index Terms—Alzheimer's disease, MRI, Deep learning, Convolutional Neural Networks (CNN), Image classification, Medical imaging, Transfer learning, Data augmentation, Neuroimaging, Dementia detection, Impairment detection

I. INTRODUCTION

IN 2020 it was reported that every 3 seconds someone develops dementia, with over 55 million people currently living with the condition worldwide. [1] This number is expected to increase to 153 million by 2050. [2] Alzheimer's disease accounts for approximately 60–70% of these cases, making it the most common form of dementia. [3] Given this growing prevalence and the continuously aging population, early detection of this disease has never been more important. Despite its importance, early diagnosis remains a challenge due to the subtlety of the symptoms, which can often be confused with normal aging or other neurological disorders.

Recent advances in medical imaging and artificial intelligence, particularly deep learning, offer promising approaches for improving Alzheimer's disease detection. Magnetic Resonance Imaging (MRI) has emerged as a widely used non-invasive tool for visualizing brain structures and identifying patterns of atrophy associated with Alzheimer's.

II. STATE OF THE ART

An extensive state-of-the-art review was performed to find the most suitable models for classifying Alzheimer's disease using MRI scans. The following results, show the six key studies found:

A. Relevant MRI Features in Alzheimer's Classification

Recent studies, including the comprehensive review by Sørensen et al. (2016) [4], have demonstrated that texture-based radiomic features extracted from MRI images provide important biomarkers for the classification of Alzheimer's

disease (AD). These features capture structural and intensity-based patterns associated with neurodegeneration, often invisible to human inspection.

- **First-order features:** Statistics like mean, variance, and skewness reflect tissue degradation.
- **Second-order (GLCM):** Contrast, energy, and homogeneity capture structural patterns, especially in the hippocampus [5].
- **Fractal and entropy metrics:** Describe cortical complexity, decreasing with disease progression [6].
- **LBP and asymmetry:** Detect microtexture changes and left-right intensity differences linked to early decline.
- **Shape and frequency features:** Identify atrophy via geometry (e.g., compactness) and texture via FFT.

B. Support Vector Machines Effectiveness on Alzheimer's MRI Classification

Support Vector Machines (SVM) have proven to be an effective machine learning approach for an Alzheimer's severity classification problem using MRI data. In a study conducted by Supriyanti et al. [7], the authors were able to develop an SVM-based system for automatically classifying Alzheimer's disease severity levels from brain MRI images, focusing on the hippocampus region.

The method involved preprocessing MRI scans and extracting important features. These features served as inputs for the classifier that distinguished between three classes based on three severity levels: normal, mild and moderate. The proposed system achieved a classification accuracy of **90.9%**, demonstrating the potential of using SVMs in detecting early diagnosis of Alzheimer's disease.

C. Comparative Analysis of XGBoost and SVM for Alzheimer's Disease Classification Using MRI

In a study by Li et al. [8], the authors utilized the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset to develop predictive models for early AD detection. They employed data preprocessing techniques, including handling missing values and outliers, followed by feature selection using Spearman correlation analysis. Three machine learning models (Random Forest, XGBoost, and SVM) were trained and evaluated. The XGBoost model outperformed the others, achieving an accuracy of 91%.

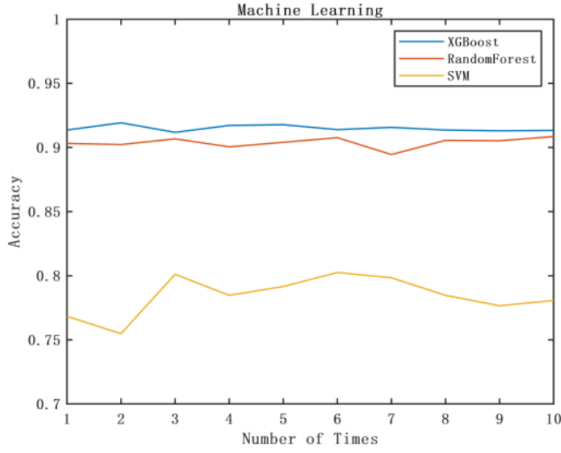


Fig. 1: Accuracy comparison of SVM, Random Forest, and XGBoost classifiers over 10 experimental runs (Image from the study by Li et al. [8]).

D. VGG16 for Alzheimer's Classification

Sharma et al. [9] applied a VGG16-based CNN model for Alzheimer's stage classification using brain MRI scans. Leveraging transfer learning, the model achieved strong results on two datasets, with up to **90.4% accuracy** and an AUC of **0.969**.

This study highlights the effectiveness of pre-trained models like VGG16 in medical imaging, especially when working with limited labeled data, and supports its inclusion in comparative evaluations.

E. EfficientNetB0 for Alzheimer's Classification

Ujilast et al. [10] utilized EfficientNetB0 to classify MRI images into four categories: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. Their model achieved an accuracy of 96.00%, demonstrating the effectiveness of EfficientNetB0 in handling classification tasks with high precision.

III. DATA DESCRIPTION, VISUALIZATION AND STATISTICAL ANALYSIS

A. Data Description and visualization

The Alzheimer's MRI dataset used in this study was obtained from Kaggle [11]. This dataset comprises a comprehensive collection of Magnetic Resonance Imaging (MRI) scans labeled across four cognitive diagnostic categories:

- Impairment
- Very Mild Impairment
- Mild Impairment
- Moderate Impairment

A few examples of each class can be seen in Figure 2.

All MRI scans in the dataset are in grayscale format and were already resized to a consistent dimension of 128×128 pixels which ensures a uniform input to the model. The intensity distributions were found to be relatively consistent across

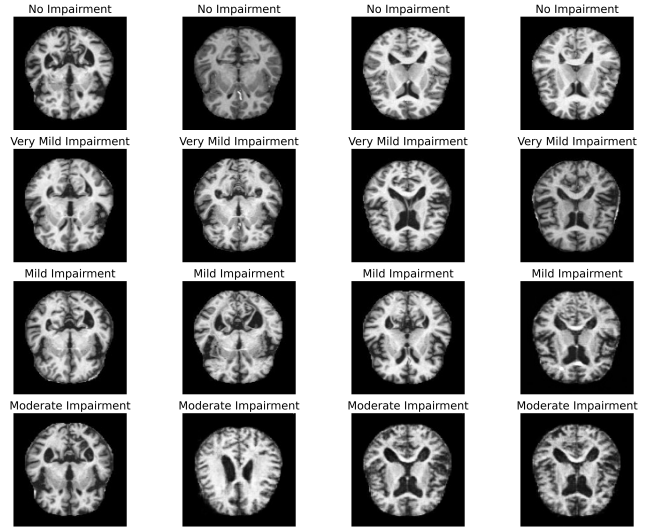


Fig. 2: Example of 4 images of each class from the training dataset

classes, indicating standardized acquisition or preprocessing steps. This uniformity minimizes the risk of introducing unwanted variability during training.

IV. STATISTICAL ANALYSIS

The original dataset consisted of a predefined training and test set which in total contained a total of 11519. The dataset creators ensured that the test set only contained real, previously unseen data, which helps provide a realistic evaluation of model generalization.

To further the development of the different models, the training set was split into two subsets: training and validation. This division allows for tuning model hyperparameters and evaluating performance without touching the final test set. The split used was of 85% and 15% for training and validation, respectively.

As shown in Figure 3, both the training and validation sets were intentionally balanced to prevent bias during the learning process. Each of the four classes comprises 25% of the data in these subsets.

However, the test set shows a significant class imbalance that mirrors a more realistic clinical distribution, where most patients are either cognitively normal or in the early stages of impairment. As seen in Table I, the "No Impairment" class alone represents 50% of the test set, while the "Moderate Impairment" class accounts for less than 0.9%. This discrepancy presents a major challenge for model evaluation, as it may lead to biased performance metrics. In particular, models may achieve high accuracy simply by predicting the majority classes, while performing poorly on underrepresented but clinically critical classes such as "Moderate Impairment".

V. DATA PREPROCESSING

Before training our machine learning models, each MRI image underwent a robust feature extraction pipeline. The

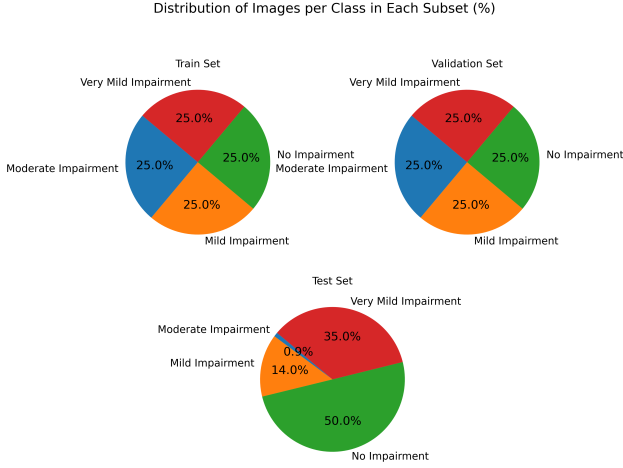


Fig. 3: Class distribution of images in the training, validation, and test subsets. Note that the training and validation sets are balanced, while the test set is highly imbalanced.

TABLE I: Number of Images per Class in Each Subset

Class	Train	Validation	Test
No Impairment	2176	384	640
Very Mild Impairment	2176	384	448
Mild Impairment	2176	384	179
Moderate Impairment	2176	384	12

types of features computed were informed by the state-of-the-art literature in Alzheimer’s imaging biomarkers (see *Section II-A*).

In particular, the following categories were extracted:

- **First-order histogram statistics:** mean, variance, skewness, and kurtosis.
- **Second-order texture metrics:** GLCM-based features (contrast, homogeneity, energy, correlation).
- **Fractal and entropy measures:** entropy and fractal dimension.
- **Asymmetry measures:** intensity and texture asymmetry across hemispheres.
- **Shape descriptors:** compactness, eccentricity, solidity of brain regions.
- **Frequency-domain features:** Fourier-based high/low frequency ratios.
- **Regional aggregation:** 3×3 image grid statistics (mean, std).

After extracting the features, the newly dataset had 112 features.

VI. MACHINE LEARNING ALGORITHMS AND RESPECTIVE RESULTS

A. XGBoost

To start, as the first machine learning algorithm, Extreme Gradient Boosting (XGBoost) was chosen due to being mentioned several times as one of the main traditional ML models in medical imaging tasks. This model is often chosen before deep learning models because it allows for a more interpretable

and computationally efficient approach, especially when working with structured, tabular data like radiomic features. The fact that it doesn’t act like a “**black box**”, making feature importance trackable, which is an important advantage in clinical settings where **interpretability** is essential. Before training the final XGBoost model, a manual grid search was performed using a validation set to tune `max_depth`, `learning_rate`, and `n_estimators`. The best combination was selected based on balanced accuracy. Early stopping and regularization `reg_alpha`, `reg_lambda` were applied to prevent overfitting.

TABLE II: XGBoost Classification Report on the Test Set (Multiclass)

Class	Precision	Recall	F1-score	Support
No Impairment (0)	0.8362	0.9016	0.8677	640
Very Mild Impairment (1)	0.7967	0.7612	0.7785	448
Mild Impairment (2)	0.8026	0.6816	0.7372	179
Moderate Impairment (3)	0.5556	0.4167	0.4762	12
Accuracy	0.8170			
Balanced Accuracy	0.6902			
Macro Avg	0.7478	0.6902	0.7149	1279
Weighted Avg	0.8151	0.8170	0.8145	1279

Despite achieving a reasonable overall accuracy of 81.70% in the multiclass setting, the model struggled to correctly classify underrepresented classes, particularly *Moderate Impairment* (Class 3), which only achieved a recall of 41.67%. This underperformance was largely due to the severe class imbalance in the test set, where Class 3 had only 12 examples out of 1279.

Given the clinical importance of distinguishing between impaired and non-impaired patients, especially when we are dealing with screening contexts, the classification problem was reframed as a binary task. The four original classes were grouped into two categories:

- **Class 0:** No Impairment (original class 0)
- **Class 1:** Any Impairment (combining original classes 1, 2, and 3)

This binary simplification aims to increase sensitivity to early signs of cognitive decline, reducing the risk of false negatives in impaired cases. A new XGBoost model was trained using the same set of features. It’s worth noting that we used a threshold of 0.35 to adjust the decision boundary of the classifier based on the predicted probabilities, rather than relying on the default threshold of 0.5. This threshold was optimized to prioritize sensitivity for the impaired class, thereby reducing the number of false negatives which is critical in early cognitive impairment detection.

TABLE III: XGBoost Binary Classification on the Test Set

Class	Precision	Recall	F1-score	Support
No Impairment	0.8709	0.8641	0.8675	640
Impaired	0.8649	0.8717	0.8683	639
Accuracy	0.8679			
Balanced Accuracy	0.8679			
Macro Avg	0.8679	0.8679	0.8679	1279
Weighted Avg	0.8679	0.8679	0.8679	1279

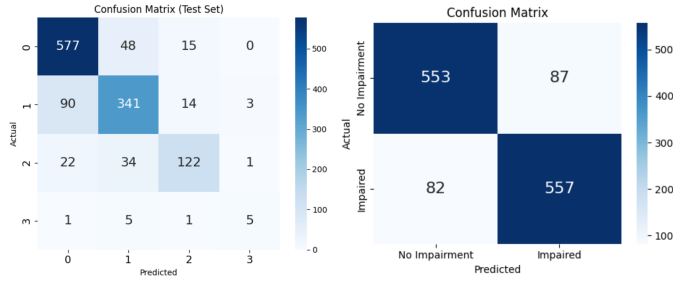


Fig. 4: Confusion matrices of the XGBoost model: on the left, the multiclass classification with four classes. The binary setup shows a slightly stronger recall for impaired cases, making it more suitable for early screening applications.

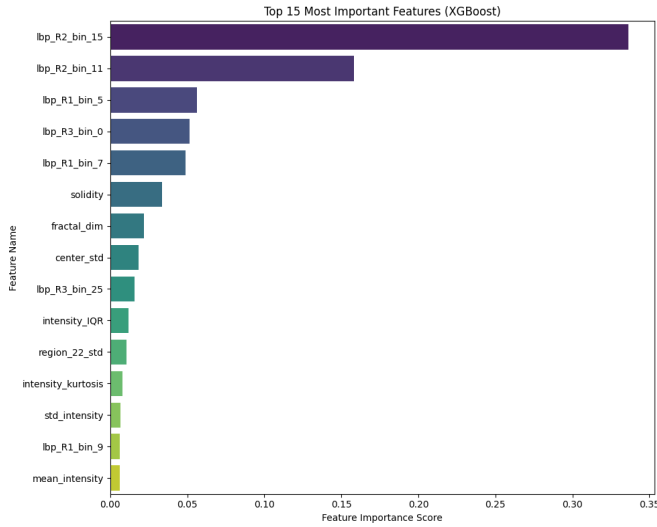


Fig. 5: Top 15 most important radiomic features ranked by XGBoost

B. Feature Importance Analysis

As shown in Figure 5, the most important features for the XGBoost model were texture-based, especially Local Binary Patterns (LBP), which are known to capture early structural changes in brain tissue linked to Alzheimer’s disease [4]. Besides that, the ‘shape’ features like fractal dimension or solidity also contributed, reflecting *cortical complexity and atrophy* [6]. Overall, the model focused on clinically meaningful patterns, aligning with prior radiomic studies in neuroimaging [12].

C. Support Vector Machine (SVM)

The Support Vector Machine (SVM) classifier was selected as the first method due to its established performance in medical imaging tasks, particularly in neurodegenerative disease classification. SVMs are very effective in high-dimensional feature spaces and where the data is limited, which makes it especially suitable for this problem where the number of data is limited.

In this study, a non-linear Radial Basis Function (RBF) kernel was used to allow the model to capture complex relationships between the extracted radiomic features and the

cognitive impairment labels. Prior to training, all features were standardized to ensure equal contribution across the different scales. Instead of relying on internal cross-validation, a dedicated hold-out validation set was used to perform a manual grid search over the key hyperparameters: the regularization parameter C and the kernel coefficient γ . This approach allowed for more transparent tuning and a clearer separation between the optimization and evaluation phases.

To mitigate the effects of class imbalance, especially the lower representation of more severe impairment classes, the SVM was trained with a balanced class weighting scheme. After selecting the optimal hyperparameters based on validation set performance, the model was retrained on the combined training and validation sets to make full use of the available data. After that a final evaluation was then conducted on the test set to assess generalization performance.

TABLE IV: Test Set Performance (4-Class Classification)

Class	Precision	Recall	F1-Score	Support
No Impairment	0.8929	0.9250	0.9087	640
Very Mild Impairment	0.8710	0.8438	0.8571	448
Mild Impairment	0.8851	0.8603	0.8725	179
Moderate Impairment	1.0000	0.6667	0.8000	12
Accuracy	0.8851			
Balanced Accuracy	0.8239			
Macro Avg	0.9122	0.8239	0.8596	1279
Weighted Avg	0.8851	0.8851	0.8845	1279

The model achieved an overall accuracy of 88.51% and a balanced accuracy of 82.39%. While the classifier performed strongly on the most represented classes, particularly No Impairment, the performance decreased on Class 3, which is the rarest class in the dataset with only 12 samples on the test set.

Specifically, although the precision for Class 3 was perfect (1.0000), the recall was only 0.6667 which indicates that almost half of the moderate impaired cases were missed. This is concerning in a medical environment where false negatives have significant clinical implications.

Given these challenges, the classification problem was reframed as a binary classification task, as previously done with XGBoost. Results can be seen in Table ??.

TABLE V: SVM Test Set Performance (Binary Classification)

Class	Precision	Recall	F1-score	Support
No Impairment	0.9232	0.8266	0.8722	640
Impaired	0.8428	0.9311	0.8848	639
Accuracy	0.8788			
Macro avg	0.8830	0.8789	0.8785	1279
Weighted avg	0.8830	0.8788	0.8785	1279

By reframing the problem into a binary classification task the model slightly improved the overall performance, with an accuracy of 89.53%. This shift in focus allowed the model to be more sensitive to signs of decline, which is particularly important in medical screening, where the cost of missing an impaired case (a false negative) can be much higher than incorrectly flagging a healthy individual.

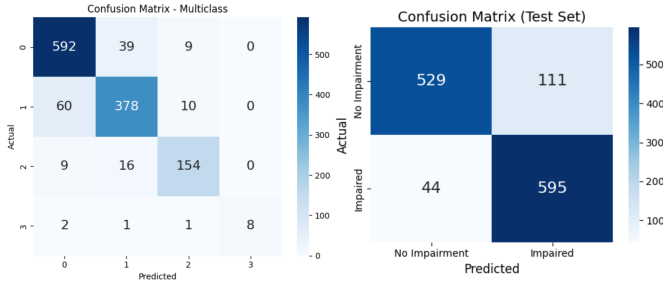


Fig. 6: Confusion matrices of the SVM model: on the left the multiclass classification with the four original classes. On the right, the binary classification. They share very similar accuracy

However, the shift to binary classification did not result in a substantial improvement in performance. The overall accuracy increased only slightly to 89.53%. This suggests that the SVM model was already effectively separating the impaired and non-impaired cases, even when the task was originally framed with four distinct classes.

As reflected in the confusion matrix (Figure 6 (right)), the model produces more false positives (111) than false negatives (44) due to the threshold of 0.35. While this might slightly reduce accuracy, it increases the likelihood that impaired individuals are flagged for further evaluation.

VII. DEEP LEARNING ALGORITHMS AND RESPECTIVE RESULTS

A. Convolutional Neural Network - VGG16

The first model used was VGG16, a convolutional neural network pre-trained on the ImageNet dataset, which contains millions of natural images, such as landscapes, objects, and animals. This pre-training enables the network to transfer knowledge to new tasks, especially those with small datasets and visual characteristics similar to natural images.

However, the fact that the network was trained exclusively on natural images represents a significant limitation. When applied to markedly different domains, such as brain magnetic resonance imaging (MRI) which presents very distinct visual characteristics knowledge transfer may be less effective. This justifies the need to perform fine-tuning of the network, adapting its parameters to the specific features of the new domain.

The architecture of VGG16 consists of 13 convolutional layers, responsible for extracting visual features from images such as edges, textures, and shapes, and 3 fully connected (dense) layers, which make decisions based on the extracted representations. The model has approximately 138 million parameters, making it powerful in terms of learning capacity, though resulting in a relatively long training time.

To streamline the process and reduce training time, the technique of Transfer Learning was employed, allowing the model to leverage previously acquired knowledge and efficiently apply it to a new classification task.

In the first phase, only the custom layers added at the end of the network were trained, while the rest of the model remained

frozen. This allowed the new layers to learn to interpret the representations already extracted by the convolutional layers of VGG16.

In the second phase, a partial unfreezing was performed, enabling the fine-tuning of some intermediate layers of the base model. This step helped to refine more specific patterns present in medical images without drastically altering the already useful weights.

In the third and final phase, a full unfreezing was applied, allowing all layers of the model to be trained. To avoid overfitting and ensure stability during training, the Early Stopping technique was used, which halts the training process once the validation loss stops improving. This ensures more precise fine-tuning without compromising the model's generalization capability.

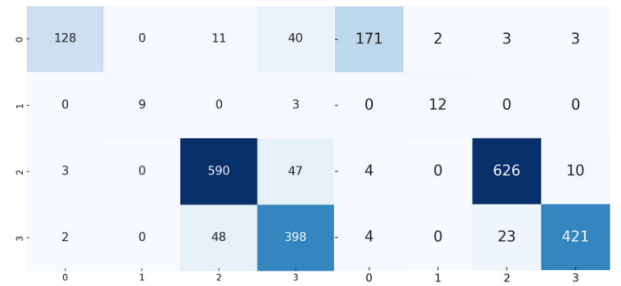


Fig. 7: Comparison of confusion matrices for the VGG16 model: partial fine-tuning (left) vs. full fine-tuning (right).

Class labels: 0 = Mild Impairment, 1 = Moderate Impairment, 2 = No Impairment, 3 = Very Mild Impairment.

TABLE VI: Model performance by class on the test set

Class	Precision	Recall	F1-score	Support
Mild Impairment	0.96	0.96	0.96	179
Moderate Impairment	0.86	1.00	0.92	12
No Impairment	0.96	0.98	0.97	640
Very Mild Impairment	0.97	0.94	0.95	448
Accuracy		0.96		1279
Macro Avg	0.94	0.97	0.95	1279
Weighted Avg	0.96	0.96	0.96	1279

The VGG16 model, after the three stages of *fine-tuning*, showed excellent performance, **achieving a final accuracy of 99.98% on the training set, 97.98% on the validation set, and 96.17% on the test set**, indicating strong generalization capability. The impact of full vs. partial fine-tuning on class-level predictions can be observed in Figure 7, which compares the confusion matrices for both strategies. The low test loss (0.1152) further reinforces the model's robustness and its ability to handle magnetic resonance imaging data, even when faced with complex and domain-specific characteristics.

Table VI shows the performance of the VGG16 model after full fine-tuning on the test set. It is evident that the model achieved excellent results across all metrics. Classes with a larger number of examples, such as *No Impairment* and *Very Mild Impairment*, obtained F1-scores of 0.97 and 0.95, respectively, demonstrating the model's effectiveness in recognizing well-represented patterns in the dataset.

Notably, the model also performed exceptionally well on the *Moderate Impairment* class, which, despite having only 12 samples, was classified with 100% precision, recall, and F1-score.

Despite the strong performance demonstrated by the VGG16 model, it takes a significant **amount of time to train**. However, it **generalizes well**. To explore potentially more efficient alternatives, we opted to compare it with a more recent and optimized architecture: **ResNet50**.

The goal of this comparison is to assess whether ResNet50 can achieve results that are comparable to or better than those of VGG16, while offering advantages in terms of training time, generalization capability, and robustness to imbalanced datasets.

B. ResNet50

This architecture is expected to deliver faster training and potentially better results, as it has fewer parameters and is capable of extracting more informative features from images. ResNet50 introduces a deeper structure and incorporates *skip connections*, which help mitigate the vanishing gradient problem commonly encountered in traditional architectures like VGG16. This model consists of 50 pre-trained layers and is much more optimized, with approximately 28 million parameters, offering greater depth and superior performance compared to VGG16.

However, despite these advantages, a limitation remains when it comes to MRI images. Although ResNet50 generally offers higher accuracy, the complexity of the Alzheimer's dataset and the degree of visual similarity between certain classes make it difficult to distinguish them, which can confuse the model.

Initially, the model was trained using all four classes: *No Impairment*, *Very Mild Impairment*, *Mild Impairment*, and *Moderate Impairment*. However, the *Moderate Impairment* class was highly imbalanced (only 12 samples), which led to a second version of the experiment, excluding that class.

In the version using all four classes, we applied a Transfer Learning approach with the ResNet50 architecture, initially freezing the convolutional base pre-trained on ImageNet and training only the newly added dense layers.

Then, we performed fine-tuning on the last layers of the network, unfreezing them for adjustment with a reduced learning rate. During training, we applied regularization techniques such as Dropout and L2 penalty (weight decay), as well as *Early Stopping* and adaptive learning rate reduction with *ReduceLROnPlateau*, to prevent overfitting and improve generalization.

Finally, the model was evaluated using the *Test-Time Augmentation* (TTA) technique, performing multiple passes with simple variations of the images such as random horizontal flipping to increase prediction robustness.

In the experiment with four classes, **the model reached a test accuracy of 84.13%**, and 73.49% with TTA. These results indicate that although the model demonstrated reasonable performance, it struggled to generalize, especially with the highly imbalanced *Moderate Impairment* class.

The drop in accuracy after applying TTA suggests that the model was sensitive to small variations in the input data, highlighting the need for more balanced datasets or more robust training strategies.

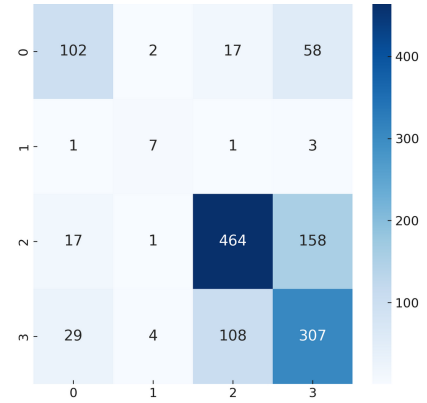


Fig. 8: Confusion matrix of the ResNet50 model including the *Moderate Impairment* class. The model shows difficulty generalizing to this minority class.

TABLE VII: Class-wise performance report on the test set

Class	Precision	Recall	F1-score	Support
Mild Impairment	0.68	0.57	0.62	179
Moderate Impairment	0.50	0.58	0.54	12
No Impairment	0.79	0.72	0.75	640
Very Mild Impairment	0.58	0.69	0.63	448
Accuracy		0.69		1279
Macro Avg	0.64	0.64	0.64	1279
Weighted Avg	0.70	0.69	0.69	1279

Since ResNet50 struggled to generalize under conditions of data imbalance and limited sample size, the *Moderate Impairment* class was removed to assess the impact of this imbalance on generalization performance.

In this new experiment, the goal was to improve the model's stability and generalization by removing the *Moderate Impairment* class, which had an extremely low number of samples and could negatively affect learning.

Initially, the convolutional base of the model was frozen. In the second phase, partial fine-tuning was applied, unfreezing the last 100 layers of the network, except for normalization layers (*BatchNormalization*), which were kept frozen to avoid training instability. Finally, a third refinement step was performed, unfreezing 50 additional layers and introducing *label smoothing* (with 10% smoothing) in the *CategoricalCrossentropy* loss function, making the model more robust to label noise.

After training, the final model was evaluated both with and without *Test-Time Augmentation* (TTA).

Following the completion of partial fine-tuning and the integration of label smoothing, the ResNet50 model was evaluated on the validation and test sets, as well as with the TTA technique. **The final model achieved an accuracy of 95.75% on the validation set, with a loss of 0.3857, and 92.27% accuracy on the test set, with a loss of 0.4437.** These

results show that the model generalized well from the training data to the test set, maintaining high performance.

However, after applying TTA with five passes using random horizontal flips the accuracy dropped to 80.51%, suggesting that, although robust in static conditions, the model remains sensitive to small variations in the input.

The detailed per-class analysis, as shown in the classification report, revealed that the *No Impairment* class achieved the best F1-score (0.86), followed by *Very Mild Impairment* (0.76) and *Mild Impairment* (0.74). The overall weighted average F1-score was 0.81, confirming consistent performance, though slightly lower in semantically similar classes (Very Mild and Mild), which are naturally harder to distinguish.

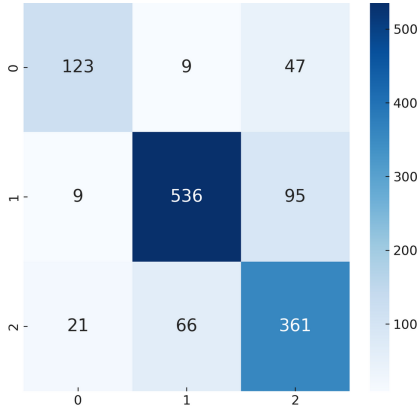


Fig. 9: Confusion matrix of the ResNet50 model trained without the *Moderate Impairment* class. More balanced performance is observed across the remaining classes, with fewer confusions between similar categories.

Class labels: 0 → Mild Impairment, 1 → No Impairment, 2 → Very Mild Impairment.

TABLE VIII: Classification report for ResNet50 model without the *Moderate Impairment* class on the test set

Class	Precision	Recall	F1-score	Support
Mild Impairment	0.80	0.69	0.74	179
No Impairment	0.88	0.84	0.86	640
Very Mild Impairment	0.72	0.81	0.76	448
Accuracy		0.81		1267
Macro Avg	0.80	0.78	0.79	1267
Weighted Avg	0.81	0.81	0.81	1267

Thus, although ResNet50 produced satisfactory results, it still relies heavily on a large volume of training data and requires considerable computational time due to its depth and number of parameters. Furthermore, as observed in the *Moderate Impairment* class, the model struggled to learn from severely underrepresented data unlike VGG16, which handles imbalanced data more effectively resulting in poor generalization and increased susceptibility to overfitting.

Given these limitations, we chose to explore the EfficientNet architecture, specifically the EfficientNetB0 model, which has gained prominence for its excellent balance between predictive performance and computational efficiency. EfficientNet introduces a novel compound scaling method that uniformly adjusts

the network’s depth, width, and input resolution, allowing the model to maintain high accuracy while remaining lightweight.

C. EfficientNetB0

Finally, we evaluated the dataset using the EfficientNetB0 architecture, a modern and optimized model with approximately 5 million parameters, which allows for significantly faster training compared to heavier models like ResNet50.

EfficientNet was designed based on a technique called *compound scaling*, which enables simultaneous and balanced scaling of the network’s depth, width, and resolution. Unlike traditional approaches that scale only one dimension, this technique results in a lighter yet highly accurate model.

The architecture employs MBConv blocks combined with *Squeeze-and-Excitation* attention mechanisms, allowing the model to capture discriminative features with fewer computational operations. This makes it ideal for scenarios with limited resources or the need for fast inference.

Training was conducted in two phases. In the first phase, the convolutional base pre-trained on ImageNet was partially frozen, and only the custom upper layers were trained. In the second phase, the entire network was unfrozen (full fine-tuning) and trained with a reduced learning rate, allowing optimization of all model parameters.

To address class imbalance, class weights were computed based on sample frequency and applied during training. The training process was supported by regularization techniques such as *Early Stopping*, dynamic learning rate reduction, and checkpointing to save the best-performing model.

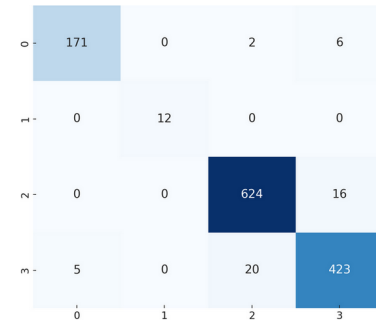


Fig. 10: Confusion matrix of the EfficientNetB0 model after full fine-tuning.

Class labels: 0 → Mild Impairment, 1 → Moderate Impairment, 2 → No Impairment, 3 → Very Mild Impairment.

TABLE IX: Performance report of the EfficientNetB0 model on the test set (with all classes)

Class	Precision	Recall	F1-score	Support
Mild Impairment	0.95	0.96	0.95	179
Moderate Impairment	1.00	1.00	1.00	12
No Impairment	0.98	0.96	0.97	640
Very Mild Impairment	0.95	0.96	0.95	448
Accuracy		0.96		1279
Macro Avg	0.97	0.97	0.97	1279
Weighted Avg	0.96	0.96	0.96	1279

TABLE X: Comparison of Models Based on Accuracy, Imbalance Sensitivity, and Training Time

Model	Train Accuracy	Validation Accuracy	Test Accuracy	Sensitivity to Imbalance	Training Time
XGBoost	99.98%	91.35%	81.70%	High	Short
SVM	99.98%	94.05%	89.52%	High	Short
VGG16	99.98%	98.24%	96.17%	Moderate	Long
ResNet50	95.48%	95.75%	92.27%	High	Long
EfficientNetB0	99.99%	98.50%	96.56%	Low	Medium

As shown in table IX, the final evaluation of the EfficientNetB0 model, trained with all four classes, **showed excellent performance** on the test set. The overall accuracy **achieved was 96.17%**, with a loss of only 0.1192, highlighting the model’s strong generalization capability.

Per-class analysis based on the classification report showed balanced and consistent results. The *Mild Impairment* class achieved an F1-score of 0.95, while *No Impairment* and *Very Mild Impairment* recorded F1-scores of 0.97 and 0.95, respectively. Notably, the *Moderate Impairment* class, despite having only 12 samples, achieved 100% recall and an F1-score of 1.00, indicating the model’s ability to correctly identify all instances of this minority class.

These results reinforce the efficiency and robustness of the EfficientNetB0 architecture, which, even with a significantly lower number of parameters, managed to match or surpass the performance of heavier architectures such as ResNet50, while requiring substantially less training time and delivering excellent inference accuracy.

VIII. GENERAL DISCUSSION AND POSSIBLE IMPLEMENTATION

The table X shows the performance of all evaluated models across key metrics.

One of the most surprising findings of this study is the distinct disconnection between model complexity and performance in the context of our imbalanced medical imaging data. In opposition to conventional expectations, larger and deeper architectures like ResNet50 did not consistently outperform older or smaller networks. In fact, EfficientNetB0, the smallest and most computationally efficient model tested, yielded the best generalization performance even outperforming the more parameter-heavy VGG16 and the theoretically superior ResNet50. This raises important questions about the prevailing assumption that deeper necessarily means better, especially in domains where data volume is limited and class imbalance is extreme.

Equally revealing was the performance of traditional machine learning models (SVM and XGBoost), which performed remarkably well despite relying on hand-crafted radiomic features. The SVM model achieved nearly 89.5% accuracy in binary classification, showing strong resilience to overfitting and class imbalance. Interestingly, it maintained relatively high recall even for minority classes, outperforming ResNet50 in this regard. XGBoost, while slightly less accurate than SVM in multiclass classification, showed better sensitivity when re-framed for binary screening.

It’s worth noting that the training on the traditional machine learning models were much quicker than the deep learn-

ing models and used way less computational resources, as expected. So trading one by the other, we end up trading interpretability and efficiency for higher accuracy.

A. Implementation

Considering the strengths and limitations of both traditional and deep learning models, a hybrid diagnostic approach appears most practical for real-world deployment.

In this setup, traditional models like XGBoost could act as an initial screening tool. Their speed, low computational requirements, and interpretability make them well-suited for use in clinical settings such as radiology triage. Trained on radiomic features, these models can effectively flag patients showing early signs of cognitive impairment—especially when tuned to favor sensitivity, minimizing missed cases.

For instances flagged as high-risk or uncertain, a second-stage evaluation using a more powerful model like EfficientNetB0 can be applied. Its ability to capture complex spatial features offers a deeper analysis, helping refine the diagnosis. Since deep learning models are more resource-intensive, limiting their use to only selected cases ensures a more efficient and scalable workflow.

IX. CONCLUSION

This study demonstrated the effectiveness of both traditional machine learning and deep learning models for Alzheimer’s diagnosis using MRI data. While SVM and XGBoost offered strong baseline performance, EfficientNetB0 achieved the highest accuracy with minimal computational cost when compared to other DL models, making a hybrid choice the go to choice in terms of deployment.

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